



BECS 21213

SOFTWARE ENGINEERING

Assignment 1 - Software Requirements Specification

Modern Residential Electricity Bill Calculator

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1. Introduction

The Modern Residential Electricity Bill Calculator is a Java application that can calculate monthly electricity bills for domestic consumers in Sri Lanka. Sri Lanka's electricity tariff structure changed in October 2025 when the Public Utilities Commission of Sri Lanka (PUCSL) introduced multi-tier slabs alongside a new Time-of-Use (TOU) rate scheme. Most consumers says that is hard to work out their own bills under the new rates. This program was built to handle both tariff plans and also for solar energy export credits under the Net Metering policy.

1.1 Purpose

This SRS document describes what the Electricity Bill Calculator does, what it needs to run, and what limits it has. It covers the functional behaviour, non-functional qualities, interfaces, and constraints of the system. The document is meant to be a reference point for the development team so that everyone is on the same page about what the software should and should not do.

1.2 Intended audience

The people who would read this document include:

- Developers working on the code, so they know what to build and test.
- Course instructors and evaluators, to check how thoroughly the requirements were analysed.
- Other team members, as a reference when making decisions or maintaining the project later.
- End users (domestic consumers), if they want to understand what the system can and cannot do.

1.3 Scope

The system takes in a consumer's name, ID, consumption figures, and solar export units, then calculates the monthly bill under either the Standard Domestic plan or the TOU plan. It prints a breakdown that includes energy charges, fixed charges, solar credits, and the net payable amount. The program runs entirely in the console. It does not have a graphical interface, does not connect to a database, and does not support online payments.

1.4 Definitions, acronyms, and abbreviations

Term	Definition
SRS	Software Requirements Specification
OOP	Object-Oriented Programming
TOU	Time-of-Use, a tariff plan with separate day, peak, and off-peak rates
PUCSL	Public Utilities Commission of Sri Lanka

Net Metering	A billing arrangement that credits consumers for solar electricity they export to the grid
kWh	Kilowatt-hour, the unit used to measure electricity consumption
Slab	A consumption range that has its own per-unit rate

Table 1: Definitions, acronyms, and abbreviations

2. Overall description

2.1 Product perspective

This is a standalone application written with Java. It does not talk to any database, web service, or external hardware or something. We built it as an OOP project to demonstrate encapsulation, inheritance, polymorphism, and abstraction in a practical settings. The tariff rates and billing logic follow the PUCSL gazette published in October 2025.

2.2 Product functions (summary)

At high level, the system does following those:

- Collect consumer details (name, ID, solar export units) from the console application.
- Lets the user pick the Standard Domestic and TOU tariff plans.
- Calculates the energy monthly charge according to the selected plan's rate structure.
- Works out the fixed charge based on the plan and consume level.
- Reduce cost from any solar export.
- Prints a formatted bill summary by the application.

2.3 User classes and characteristics

There is only one type of user: a domestic electricity consumer (or a billing clerk acting on their behalf) who wants to find what their monthly bill would be. The user needs to be comfortable typing into a command line and should have their monthly consumption handy. No technical background needed, basic computer use is expected.

2.4 Operating environment

- Runs on any computer with Java Runtime Environment (JRE) 8 or higher.
- Works on Windows, macOS and also with Linux.
- Uses the command-line console for both input and output of the application.
- Does not need network access or database or anything else.

2.5 Constraints

- Only covers residential domestic tariffs. Industrial, commercial, and religious institution categories are not included.
- Tariff rates are written directly into the TariffRate class, so changing them means editing the source code and recompiling.
- Nothing is saved between runs. Once the program closes, all the data from that session is gone.
- All interaction happens through a text console, with no GUI.

2.6 Assumptions and dependencies

- The tariff rates used are the ones gazetted by PUCSL on 15 October 2025.
- The user is expected to enter correct consumption and solar export values.
- JDK or JRE version 8 or above must already be installed on the machine.
- The solar export credit rate is assumed to stay at LKR 25.00 per kWh.

3. Functional requirements

Below are the functional requirements, each with a unique ID so they can be traced through development and testing.

ID	Requirement	Description
FR-01	Consumer data input	The system shall ask the user to enter the consumer's name, consumer ID, and solar export units (in kWh).
FR-02	Tariff plan selection	The system should let the user choose to switch the Standard Domestic plan and the TOU plan.
FR-03	Standard consumption input	When the user picks Standard Domestic, the system should ask for total monthly consumption units in kWh.
FR-04	TOU consumption input	When the user picks TOU, the system shall ask for day, peak, and off-peak consumption values separately (all in kWh).
FR-05	Tiered energy charge calculation	For Standard Domestic users, the system shall calculate energy charges with the following slab rates: LKR 4.50 for 0-30 units, LKR 8.00 for 31-60 units. If consumption exceeds 60 units, the rates switch: LKR 12.75 for the first 60 units, LKR 18.50 for units 61-90, and LKR 24.00 for anything above 90.
FR-06	TOU energy charge calculation	For TOU users, the energy charge is: (Day kWh * 35.00) + (Peak kWh * 67.00) + (Off-Peak kWh x * 21.00).

FR-07	Fixed charge determination	The system should add a fixed charge: LKR 80.00 if Standard and consumption is 30 units or more less, LKR 210.00 if Standard and consumption exceeds 30, or LKR 2,100.00 for all TOU consumers.
FR-08	Solar credit calculation	Solar credit is calculated as: solar export units * 25.00 per kWh.
FR-09	Net bill computation	Net payable = (Energy Charge + Fixed Charge) - Solar Credit.
FR-10	Bill display	The system should print a bill showing the consumer name, ID, tariff type, energy charge, fixed charge, solar credit, and net payable amount in a proper layout.
FR-11	Invalid input handling	If the user types are invalid tariff plan choice, the system should print an error message and exit the program in a proper way.

Table 2: Functional requirements

4. Non-functional requirements

ID	Category	Description
NFR-01	Performance	Bill calculation and display should complete within 1 second after the user finishes entering inputs.
NFR-02	Usability	Console should be clear enough for someone with basic computer skills can handle the system without reading a separate manual.
NFR-03	Reliability	The system must give mathematically correct results for every valid set of inputs, matching the defined tariff rates exactly.
NFR-04	Portability	The application should run on any platform that has JRE 8 or above, without needing code changes.
NFR-05	Maintainability	All tariff rates are kept in the TariffRate class. If rates change, a developer only needs to update that one class.
NFR-06	Security	The system does not save or send any personal data. Everything the user types is held in memory and thrown away when the program ends.
NFR-07	Availability	Since this is a local application, it is available any time the computer is on and Java is installed. There is no server to go down.

Table 3: Non-functional requirements

5. External interface requirements

5.1 User interfaces

Everything happens through the console. The program reads input using Java's Scanner class (from System.in) and prints output to System.out. The bill itself is laid out with labelled rows, separator lines, and the monetary values aligned to the right so they are easy to read.

5.2 Hardware interfaces

There are no special hardware requirements. A standard computer with a keyboard and a display is all that is needed. The program does not use printers, barcode scanners, smart meters, or any other peripheral device.

5.3 Software interfaces

The only software dependency is JRE 8 or above. The code uses `java.util.Scanner` from the standard library and nothing else. There are no third-party libraries, no frameworks, and no external API calls.

6. System constraints

6.1 Technical constraints

- Written in Java, so the JRE is required to run it.
- Tariff rates are hard-coded. There is no way to load them from a file or database at runtime.
- The program handles one bill at a time. You have to run it again for the next consumer.
- If the user types a letter where a number is expected, the program may throw a runtime exception rather than handling it gracefully.

6.2 Regulatory constraints

- The rates match the PUCSL gazette dated 15 October 2025. If PUCSL changes the rates, someone has to update the source code.
- The solar export credit rate follows Sri Lanka's current Net Metering policy.

6.3 Environmental constraints

- Console only, no GUI or web interface.
- No data is saved anywhere. When the program exits, everything from that session is lost.
- Built for single-user use. Multiple people cannot use it at the same time.

7. Appendices

7.1 Use case summary

The table below lists the main use cases. There is only one actor (the consumer or billing clerk), and all interaction goes through the console.

UC ID	Use case	Actor	Description
UC-01	Enter consumer details	Consumer	The user types in their name, ID, and solar export units.
UC-02	Select tariff plan	Consumer	The user picks either Standard Domestic or TOU.
UC-03	Enter consumption data	Consumer	The user enters consumption values for the chosen plan.
UC-04	View bill summary	Consumer	The system prints the calculated bill with a full breakdown.

Table 4: Use case summary

7.2 Class overview

The table below lists every class and interface in the project and what each one does.

Class / Interface	Description
ElecBillCalculator	The main class. Contains the main() method, handles user input, and ties everything together.
TariffRate	Holds all the tariff rate values (slab rates, TOU rates, fixed charges, solar export rate) as private fields with getter methods.
Consumer (abstract)	Base class for all consumer types. Stores the consumer ID, name, and solar export units. Declares abstract methods for energy charge and fixed charge calculation.
StandardConsumer	Extends Consumer. Calculates the energy charge using tiered slabs based on total monthly consumption.
TOUConsumer	Extends Consumer. Calculates the energy charge using separate day, peak, and off-peak rates.
BillingService (interface)	Defines two methods: calculateTotalBill() and printBill(). Any billing engine must implement these.
ResidentialBillingEngine	Implements BillingService. Takes a Consumer and a TariffRate, works out the final bill, and prints it.

Table 5: Class and interface overview

7.3 Glossary

Term	Definition
Domestic consumer	A household electricity user billed under the standard residential tariff.
Tiered slab	A range of consumption (e.g. 0-30 kWh) with its own per-unit rate. Higher slabs cost more.
Time-of-Use (TOU)	A tariff plan that charges different rates for electricity used during day, peak, and off-peak hours.
Net Metering	An arrangement where consumers with solar panels get a credit on their bill for electricity they send back to the grid.
Fixed charge	A flat monthly fee that gets added to the bill regardless of how much electricity was used.
Energy charge	The part of the bill that depends on how many units were consumed and which rate applies.

Table 6: Glossary